Santhosh Naidu Puppala

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PROFESSIONAL SUMMARY

Enterprise Software Engineer with 8+ years of experience and 2-3 years of focused hands-on GenAI engineering across RAG, LangGraph, MCP, AI agents, LLM applications, and workflow automation. Experienced building practical AI systems for banking and healthcare environments where reliability, governance, auditability, and business workflow integration matter as much as model output quality.

- Production GenAI systems: RAG applications, agentic workflows, LangGraph orchestration, MCP tool integrations, LLM-powered support automation, and FastAPI services.
- Enterprise engineering foundation: Java, Python, SQL, ETL, CI/CD, Docker, Kubernetes, Terraform, AWS, Azure, GCP Vertex AI, Databricks, and PySpark.
- Responsible AI and reliability: grounded retrieval, hallucination controls, output validation, fallback escalation, audit trails, prompt/version tracking, and observability metrics.
- Domain experience in financial services, healthcare data operations, telecom PLM, enterprise data platforms, and production support workflows.
- Strong fit for GenAI Engineer, LLM Application Engineer, Agentic AI Engineer, AI Platform Engineer, and Python AI Backend roles.

TECHNICAL SKILLS

GenAI / LLM Applications: LangChain, LangGraph, RAG, GraphRAG concepts, Prompt Engineering, OpenAI API, Gemini, Vertex AI, Hugging Face, LangSmith, LLMOps

Agentic AI: Multi-agent workflows, planner/executor/evaluator patterns, tool calling, MCP / FastMCP, memory, task routing, human approval patterns, fallback handling

Backend / APIs: Python, FastAPI, Java, REST APIs, SQL, PostgreSQL, SQLite, structured outputs, Pydantic-style schemas, JSON contracts

Data / Retrieval: ChromaDB, Pinecone, OpenSearch, embeddings, semantic search, hybrid retrieval, metadata filtering, Databricks, PySpark, Azure Data Factory

Cloud / DevOps: AWS ECS/EKS/CodeBuild/CodePipeline, Azure, GCP Vertex AI, Docker, Kubernetes, Terraform, CI/CD, GitHub, Jenkins

Observability / Governance: Prometheus, Grafana, ELK, CloudWatch, tracing, evaluation metrics, confidence scoring, groundedness checks, audit logging

PROFESSIONAL EXPERIENCE

Senior AI/GenAI Engineer - Agents & LangChain | PNC Bank (via DVI Technologies Inc.), Pittsburgh, PA | Oct 2024 - Present

- Designed and deployed GenAI solutions using Vertex AI Gemini, OpenAI API, LangChain, and LangGraph to automate incident triage, production support, and operational workflows for enterprise banking applications.
- Built multi-agent LangGraph workflows with planner, executor, evaluator, and memory components to break complex production incidents into automated sub-tasks and reduce manual support effort.
- Integrated SQL execution, log search, NAS path validation, job-status lookup, and configuration inspection through MCP-style tool interfaces, allowing agents to interact with governed enterprise systems.
- Built RAG pipelines over 500+ runbooks and support documents using vector search, metadata filters, and citation-grounded retrieval to improve issue diagnosis and reduce reliance on manual documentation search.
- Implemented session memory, correlation IDs, audit trail patterns, and workflow state tracking for reliable multi-turn GenAI interactions in regulated banking support environments.
- Developed reusable prompt libraries for incident analysis, RCA generation, error classification, and support response drafting, with validation patterns to reduce hallucination and unsupported answers.
- Applied Responsible AI controls including confidence scoring, grounded retrieval checks, output validation, fallback escalation, and human review for low-confidence responses.
- Designed FastAPI-based AI services with CI/CD, Docker/Kubernetes deployment patterns, cloud integration, and observability for tool calls, retrieval quality, latency, and agent state transitions.
- Mentored junior engineers on prompt design, RAG architecture, LangGraph workflows, evaluation patterns, and production GenAI engineering practices.

GenAI Engineer / AI Data Engineer | SysArch Inc. / Zimmer Biomet, McLean, VA | Jun 2024 - Oct 2024

- Built a GenAI-assisted knowledge platform using Python, LangChain, and RAG to retrieve pipeline documentation, classify ETL failures, and generate structured recommendations for healthcare data teams.
- Developed an AI Data Quality Assistant that reviewed validation failures, schema mismatches, reconciliation issues, and pipeline anomalies, then produced actionable summaries for data operations teams.
- Designed enterprise ETL pipelines using Azure Data Factory, Databricks, and PySpark with metadata-driven transformations and data-quality validation for healthcare analytics workloads.
- Implemented LLM-powered semantic search across data dictionaries, schema definitions, and pipeline documentation to reduce manual investigation time for engineering teams.
- Built automated reconciliation workflows using PySpark and Databricks to validate source-to-target counts, null checks, referential integrity, and downstream reporting readiness.

- Created structured prompt templates for ETL error reports, root-cause summaries, impacted table lists, and remediation steps while respecting healthcare data governance constraints.

Sr. Software Engineer | Qentelli, Hyderabad, India | Jan 2022 - Jan 2023

- Designed and delivered cloud-native enterprise applications on AWS using ECS, Elastic Beanstalk, CodeBuild, Docker, and Kubernetes, improving deployment repeatability across environments.
- Built CI/CD pipelines with AWS CodeBuild and CodePipeline for Java and Python applications across development, staging, and production environments.
- Provisioned cloud infrastructure using Terraform including VPCs, subnets, IAM, S3, RDS, and security groups to support repeatable and secure deployments.

Senior Software Engineer | HCL Technologies (Client: Nokia), Bengaluru, India | Apr 2019 - Jan 2022

- Served as Informatica PowerCenter ETL middleware engineer for Nokia enterprise data platforms, designing mappings and workflows processing millions of records across 26+ downstream applications.
- Supported ENOVIA version upgrades by assessing downstream data impact, updating transformation logic, validating pipelines, and ensuring zero production data loss.
- Collaborated with business, data, and application teams to troubleshoot production issues, improve batch reliability, and maintain enterprise data flow quality.

Project Engineer | Wipro Limited (Client: Nokia), Bengaluru, India | Jun 2016 - Mar 2019

- Worked on Nokia ENOVIA PLM platform for BOM management, engineering change management, product configuration, and enterprise data workflows.
- Developed Java backend components and TCL/MQL automation scripts for PLM processes, upgrade validation, regression testing, and compatibility fixes.

PROJECTS

Enterprise GenAI Support Assistant | github.com/SanthoshPuppala94/enterprise-genai-support-assistant

- Built a production-style GenAI support assistant with LangGraph supervisor routing to SQL, Document/RAG, Log Troubleshooting, and Letter Explanation agents.
- Implemented FastMCP tools, FastAPI /chat endpoint, SQLite memory, citation-grounded RAG, SQL mutation prevention, hallucination controls, environment-based secrets, and pytest coverage.

Stack: Python, LangGraph, LangChain, FastAPI, FastMCP, SQLite, RAG, OpenAI API, pytest

MCP DeepAgent Workbench | github.com/SanthoshPuppala94/mcp-deepagent-workbench

- Built an agent framework demonstrating MCP server design, LangGraph lifecycle control, persistent memory, tool adapters, and explicit state transitions from plan to synthesis.
- Implemented decorated MCP tools/resources, LangChain StructuredTool adapters, evidence-limited fallbacks, citation checks, FastAPI services, and pytest coverage.

Stack: Python, LangGraph, LangChain, FastAPI, FastMCP, SQLite, StructuredTool, pytest

EDUCATION

Master of Science, Computer Science - Data Science | Wilmington University, New Castle, DE | May 2024 | GPA: 3.4

ADDITIONAL STRENGTHS

- Hands-on builder of GenAI proof-of-concepts that demonstrate enterprise architecture, testing, API contracts, guardrails, and deployment readiness.
- Comfortable translating business support workflows into AI-assisted systems that combine LLM reasoning with tools, APIs, logs, documents, databases, and human approval paths.